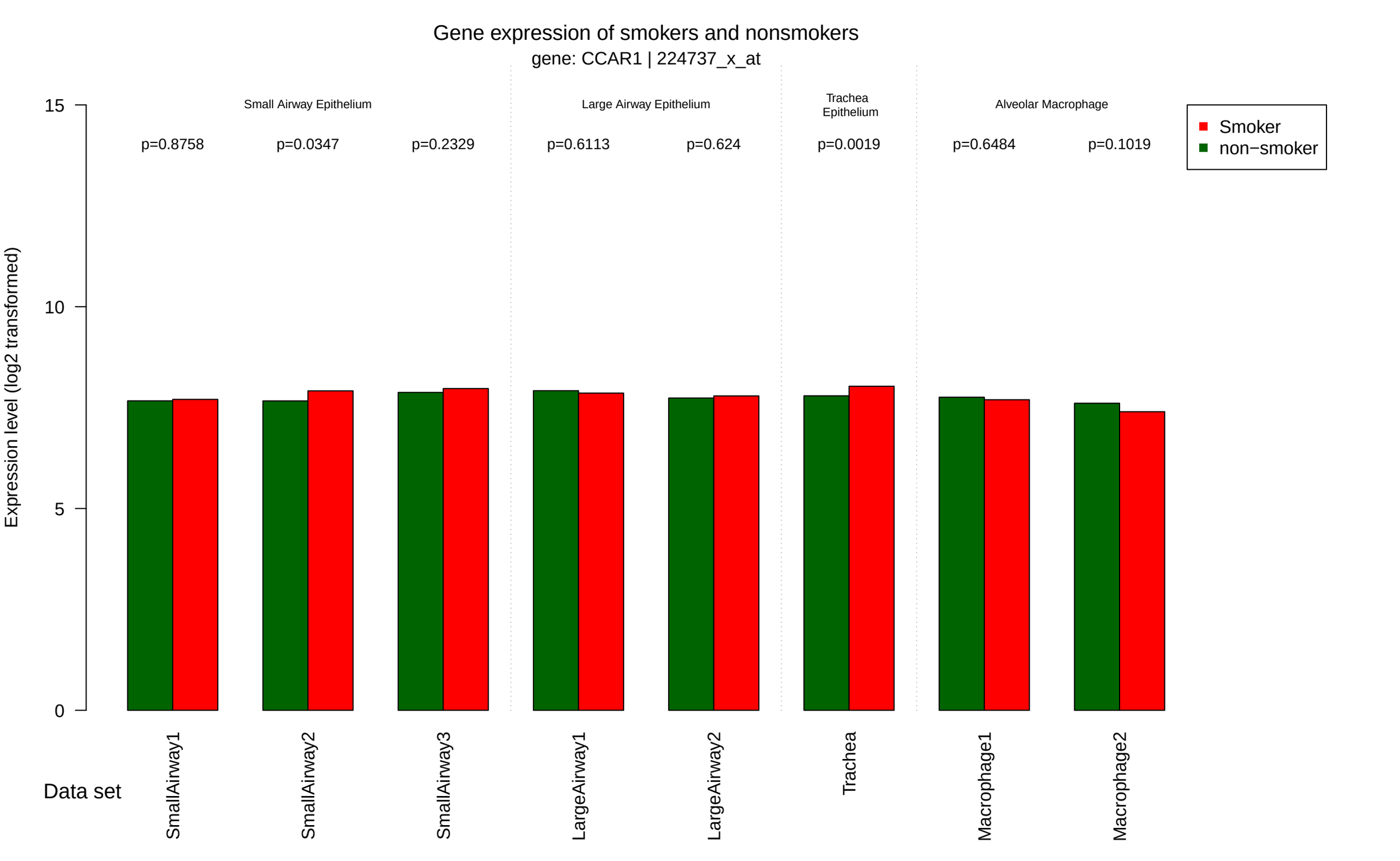
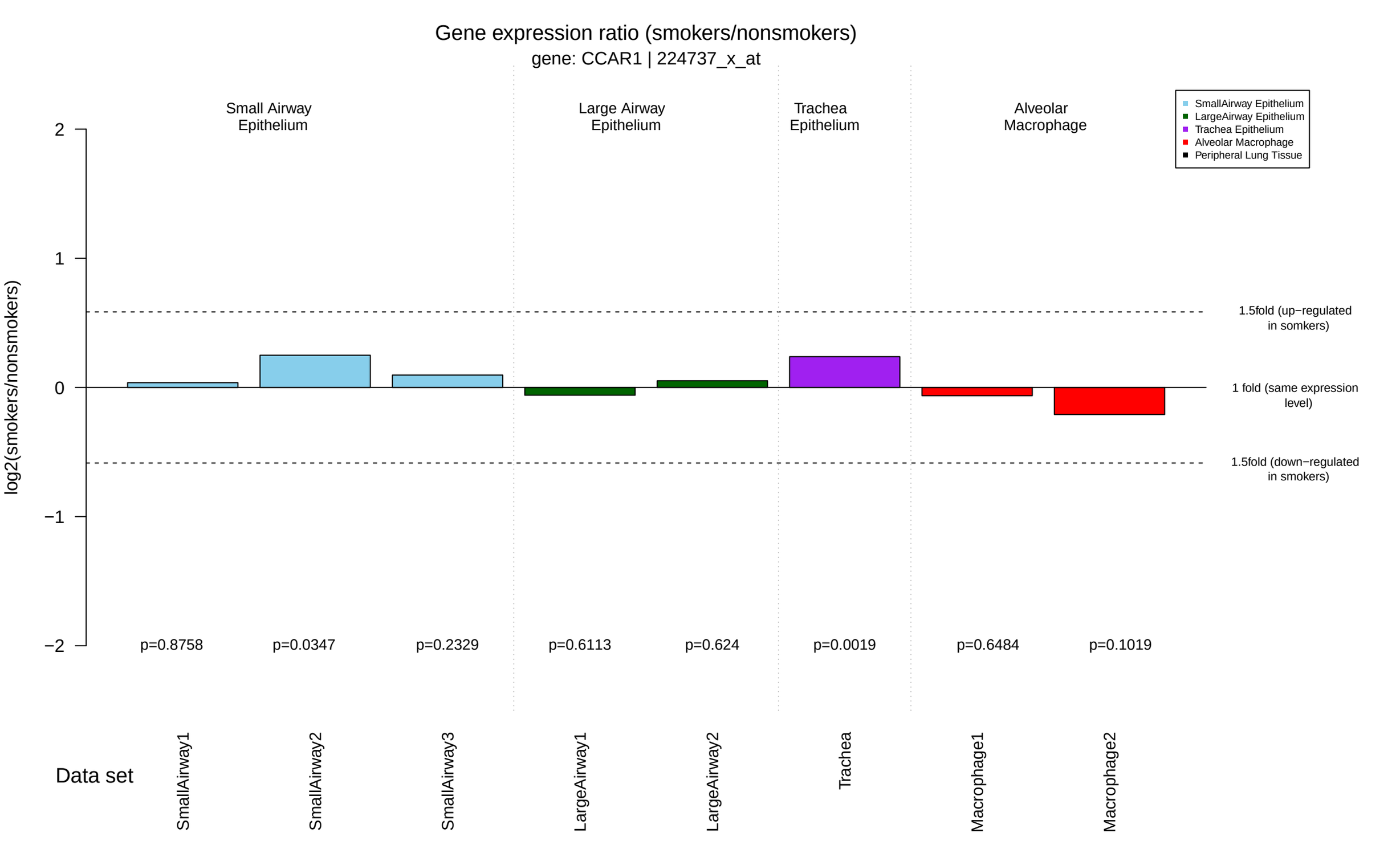
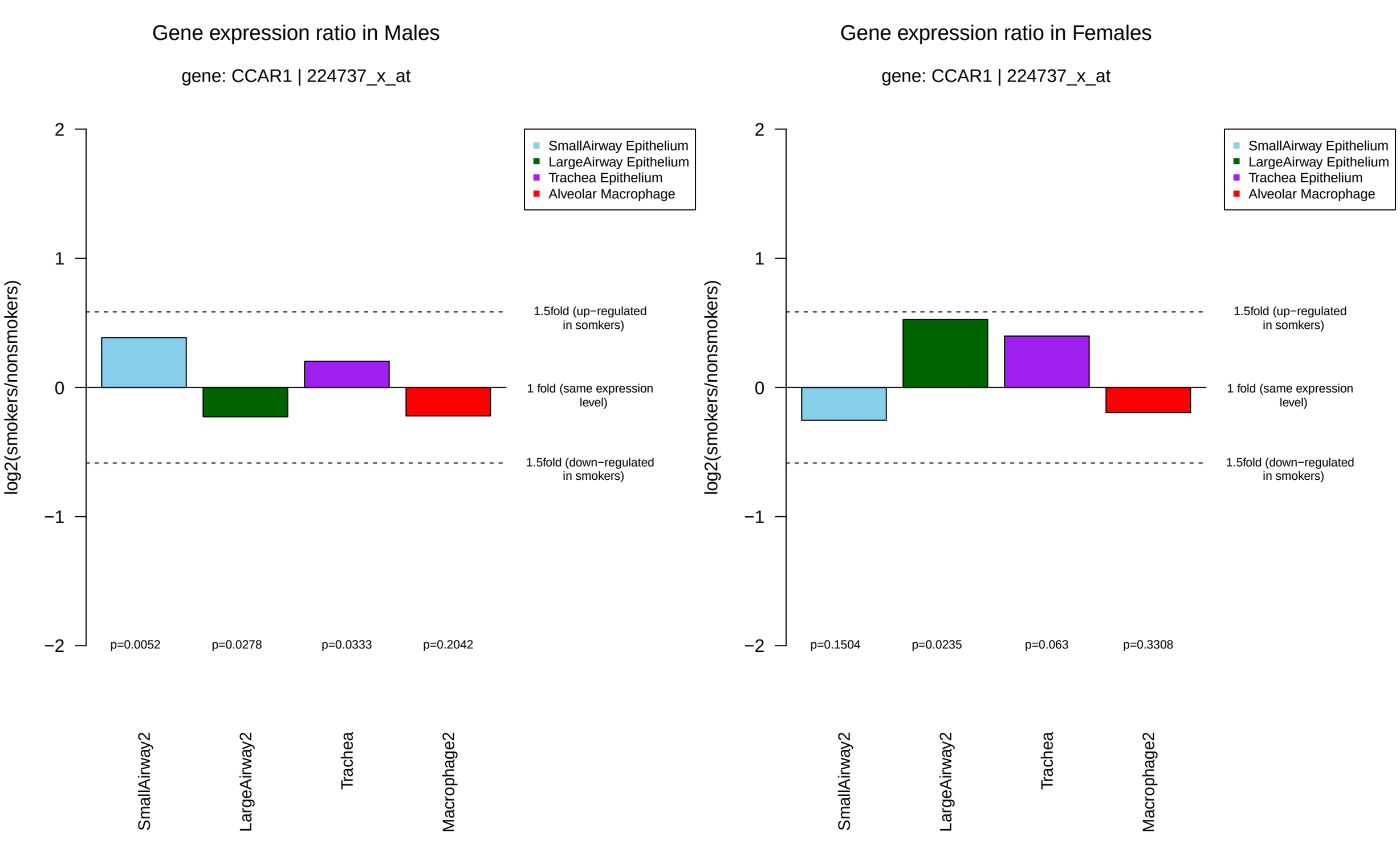
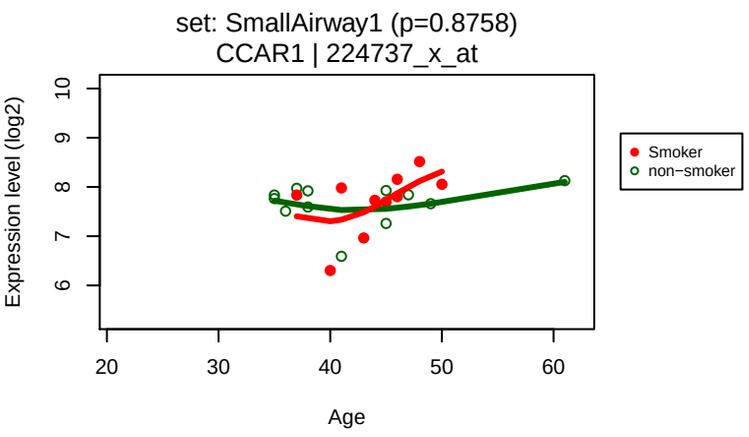
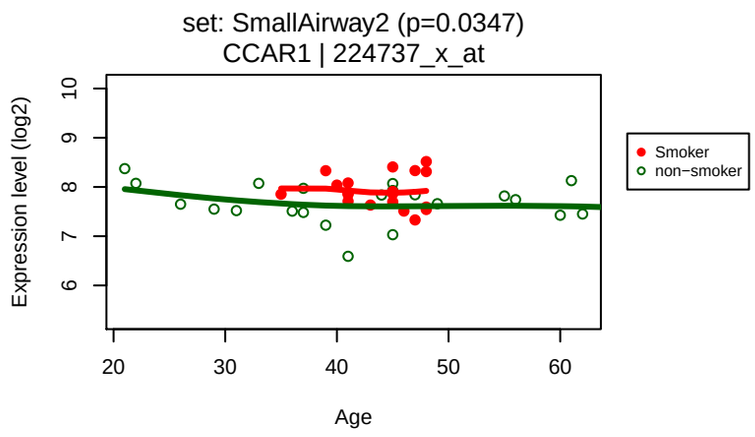
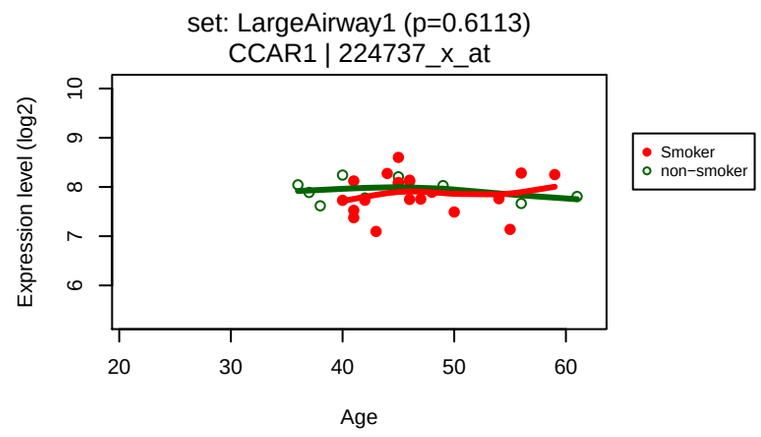

Gene expression of smokers and nonsmokers
gene: CCAR1 | 224737_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8758
p=0.0347
p=0.2329
p=0.6113
p=0.624
p=0.0019
p=0.6484
p=0.1019
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

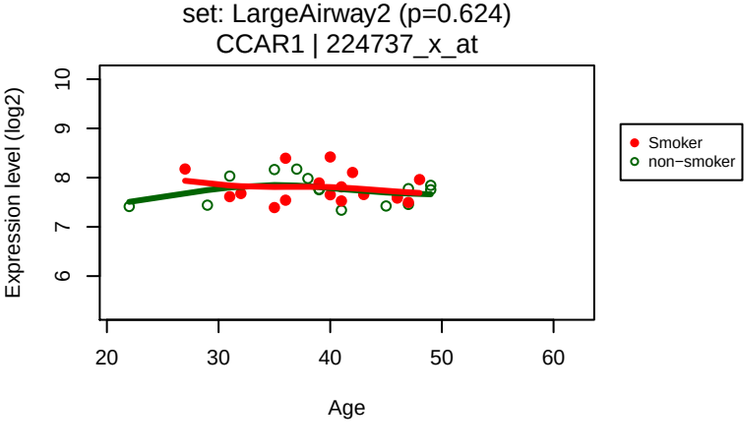

Gene expression ratio (smokers/nonsmokers)
gene: CCAR1 | 224737_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8758
p=0.0347
p=0.2329
p=0.6113
p=0.624
p=0.0019
p=0.6484
p=0.1019
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

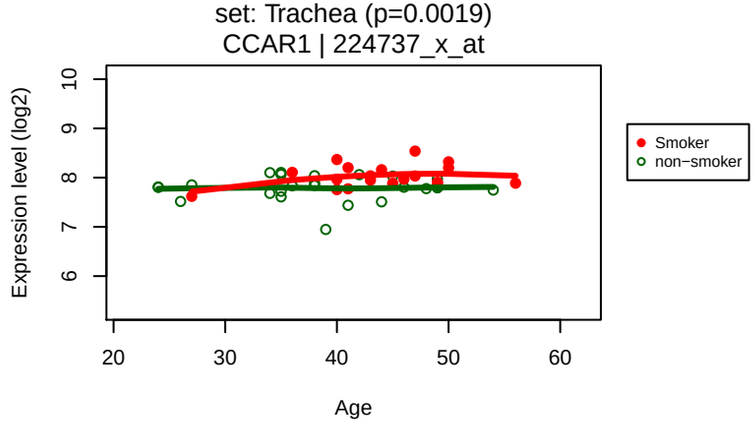

Gene expression ratio in Males
gene: CCAR1 | 224737_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0052
p=0.0278
p=0.0333
p=0.2042
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CCAR1 | 224737_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1504
p=0.0235
p=0.063
p=0.3308
SmallAirway2
LargeAirway2
Trachea
Macrophage2

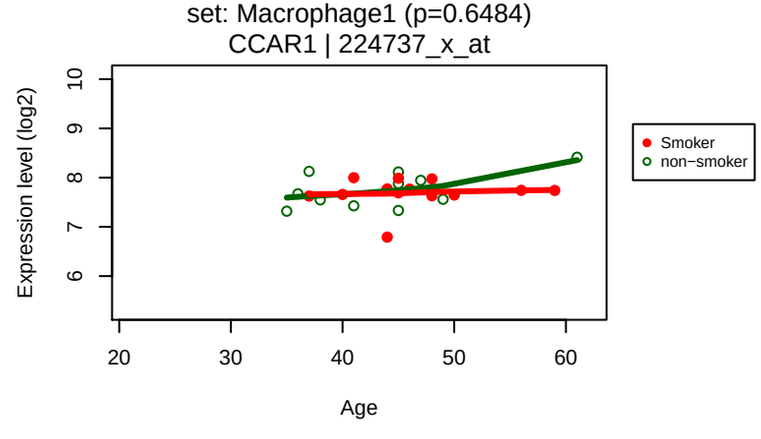

set: SmallAirway1 (p=0.8758)
CCAR1 | 224737_x_at
Expression level (log2)
Age
Smoker
non−smoker

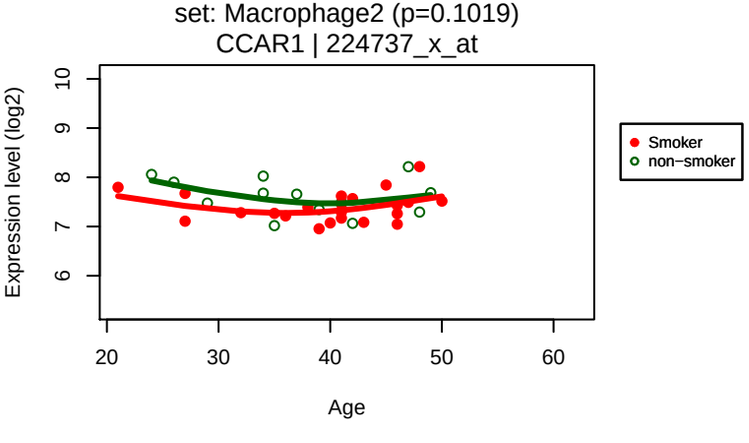

set: SmallAirway2 (p=0.0347)
CCAR1 | 224737_x_at
Expression level (log2)
Age
Smoker
non−smoker

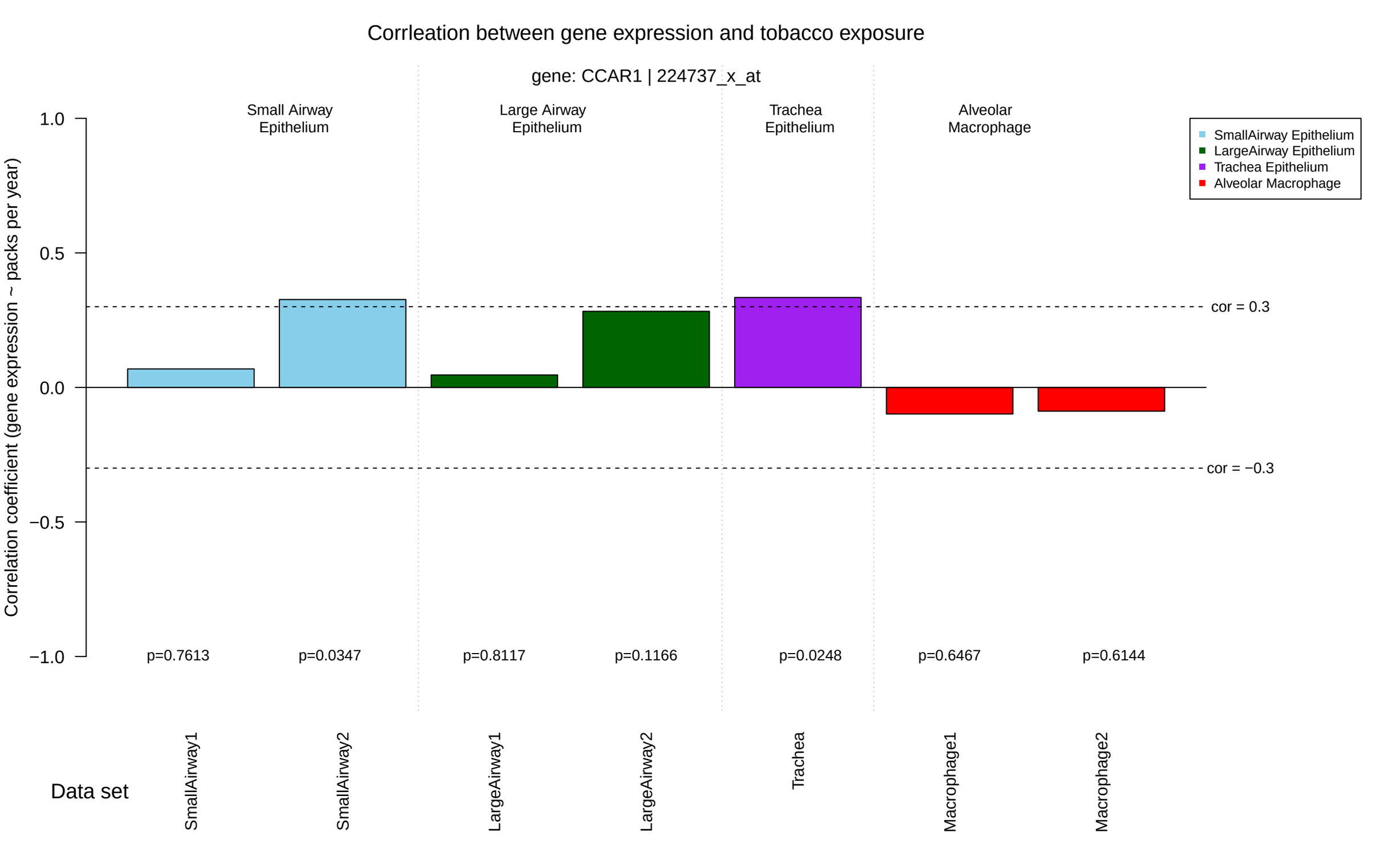

set: LargeAirway1 (p=0.6113)
CCAR1 | 224737_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.624)
CCAR1 | 224737_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0019)
CCAR1 | 224737_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6484)
CCAR1 | 224737_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1019)
CCAR1 | 224737_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CCAR1 | 224737_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7613
p=0.0347
p=0.8117
p=0.1166
p=0.0248
p=0.6467
p=0.6144
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2